What Factors Are Most Associated with Football Player Market Value?

DATA 400 Individual Mini-Project

Niloy Saha · Dickinson College · Fall 2026

Question and motivation

RESEARCH QUESTION

Which player characteristics and performance statistics are most strongly associated with a football player's market value?

- Player valuations vary enormously, from tens of thousands to hundreds of millions of euros.
- Players with similar output can carry very different values.
- Goal: describe which characteristics move with value, not predict a "true price".

Market value ≠ transfer fee

Transfermarkt's estimate at a point in time: not a price paid, not a true value.

WHO CARES

Fans and journalists who quote values, analysts and recruiters who use them, players and agents affected by them.

Data: five seasons of Transfermarkt history

32,046
player-seasons

13,511
distinct players

14
domestic European leagues

5
seasons, 2020/21 to 2024/25

One row = one player in one season

- Source: David Caribou's public transfermarkt-datasets (CC0)
- Match records summed into season totals
- A player appears once for each season played

Main variables

Age	Position
Minutes	Goals
Assists	Per-90 rates
League	Market value

From 1.9 million match rows to one analytical table

Match appearances → Games: season and league → Sum per player-season → Player details (age, position) → Season-end valuation → **Clean player-season table**

450 minutes

Per-90 rates only for players with at least 450 minutes (five full matches). 1 goal in 1 minute = 90 per 90.

July 31 rule

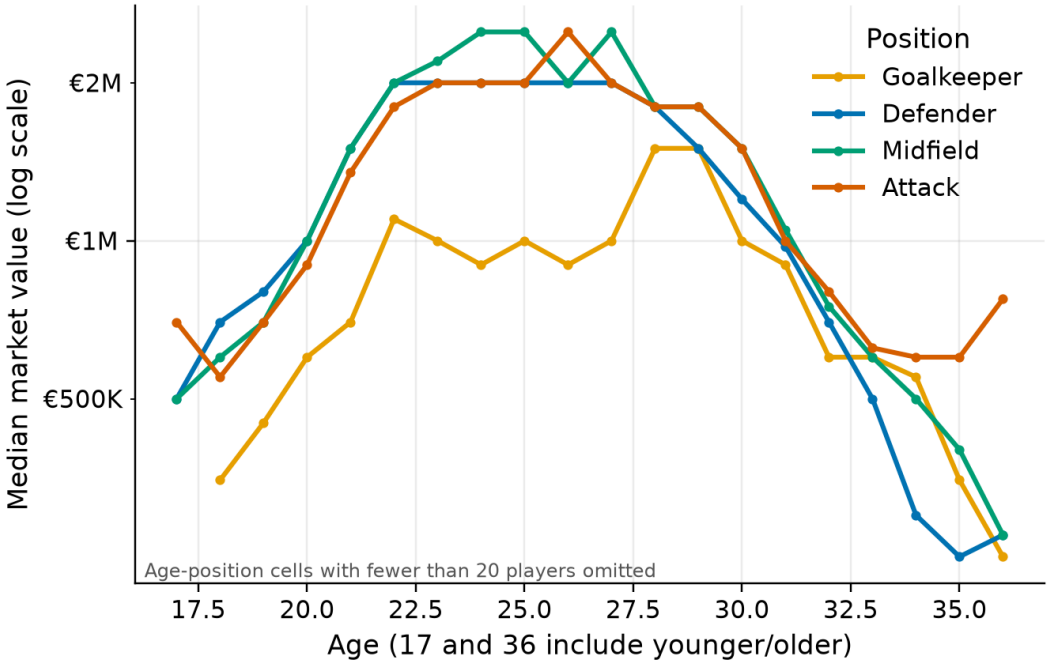
Season value = latest Transfermarkt valuation on or before July 31 after the season.

365 days

Valuations more than a year old at that date are excluded, so no stale values.

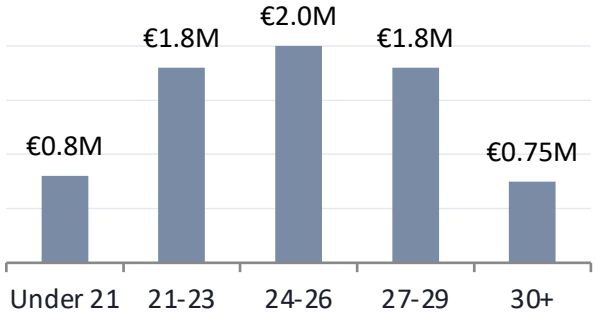
32,966 → 32,046 player-seasons: 920 rows (2.8%) excluded for missing position or birth date, missing valuations, or valuations more than a year old. Players changing clubs keep one row, labelled by the club where they played most.

Age matters, but not in a straight line



Median by age, one line per position; value axis is log-scaled (each step ×10).

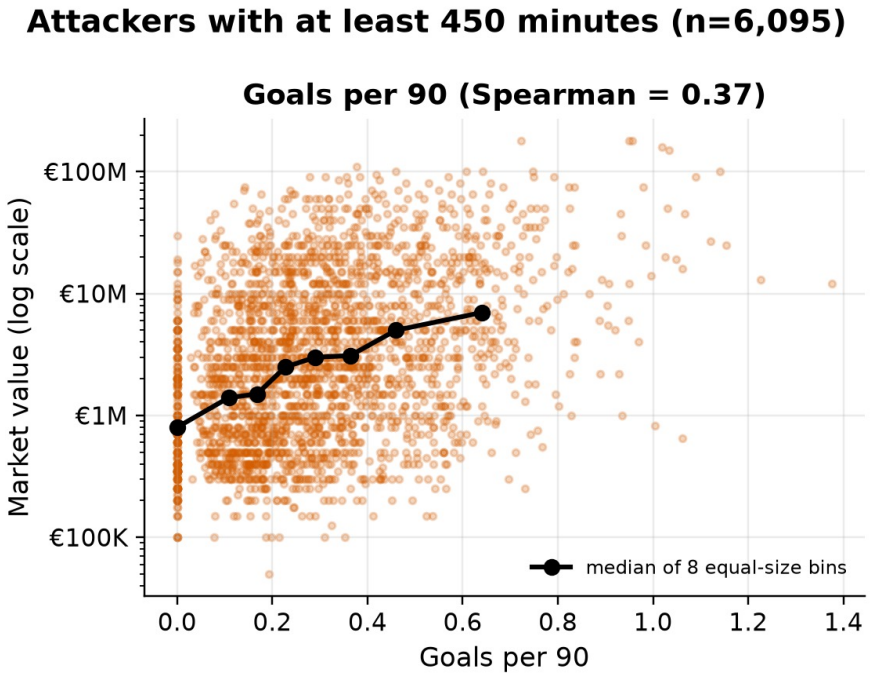
Median value by age group (all players)

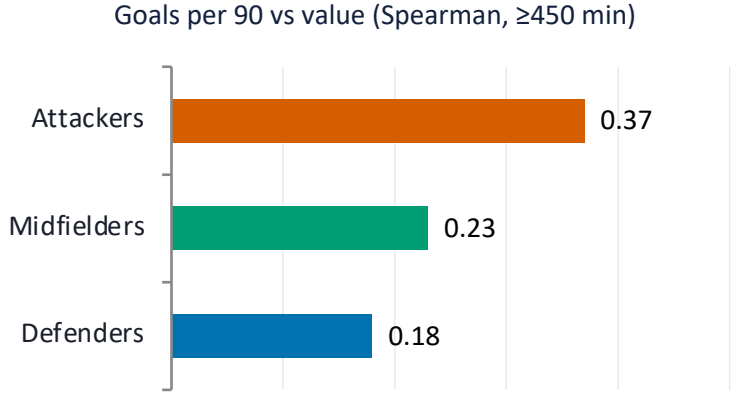


Spearman(age, value) = -0.03

Near zero, yet age clearly matters: value rises, peaks at 24–26, then falls. A single correlation hides the hump.

The same statistic means different things by position

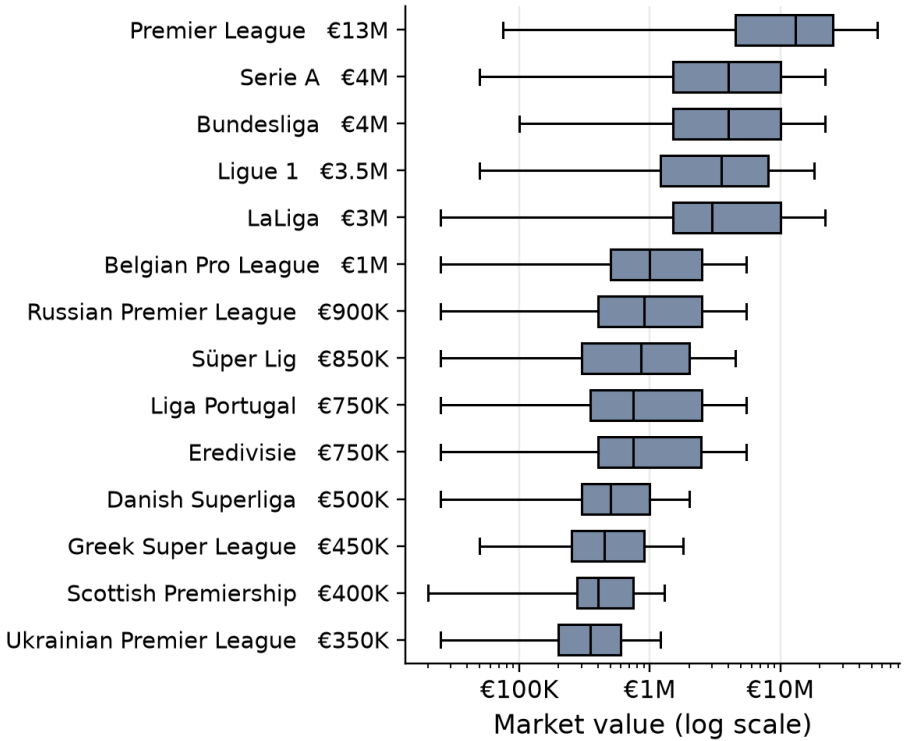




Goals track value most for attackers. That does not make goals a measure of defender quality: the data has no tackles, saves or other defensive statistics.

Dots: random sample of 2,500 attacker-seasons. Median line and correlation use all 6,095 eligible attackers.

League context shows the largest group differences



€13M

Premier League median value

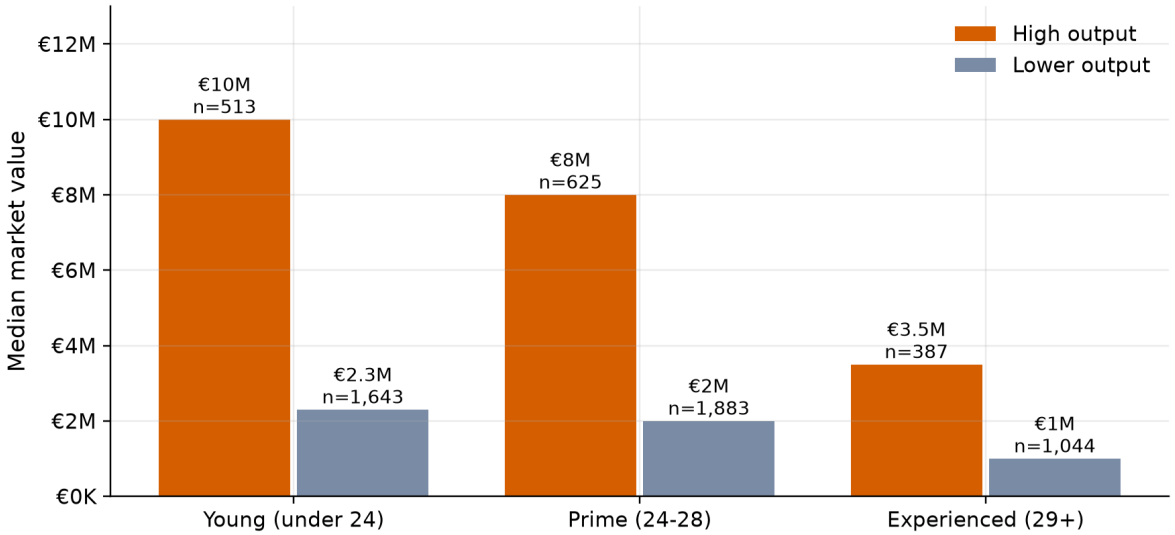
€4.5M vs €600K

top-five leagues vs the other nine (pooled medians)

Context, not cause

Leagues differ in money, clubs, player pool and exposure. About 35% of the variation in (log) value lies between leagues, vs about 1% between positions.

Young, high-output attackers: highest median values



€10M

young, high output

€8M

prime age, high output

€3.5M

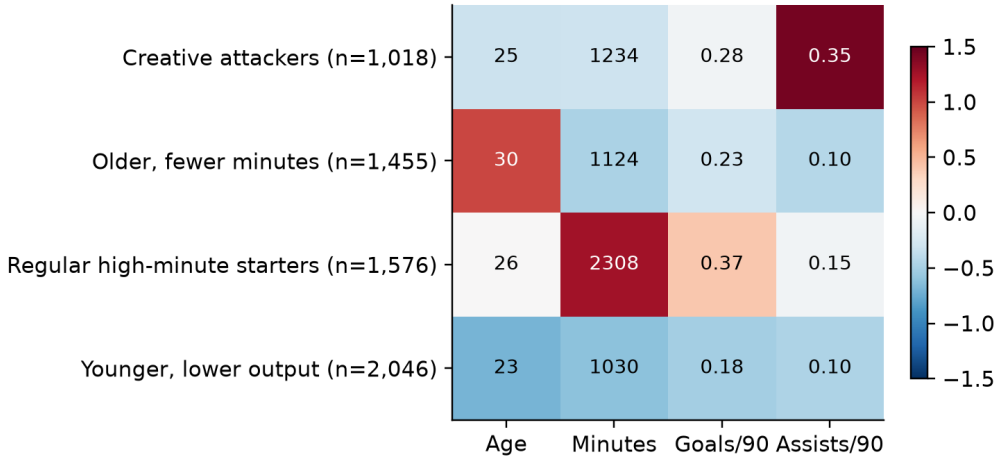
experienced, high output

Attackers with ≥450 minutes; high output = top quarter of goals + assists per 90 (≥ 0.59). Same ordering inside the top-five leagues (€31M vs €25M) and outside them (€5M vs €4.5M). Descriptive only: potential, contracts and club level are not in the data.

Do attackers form distinct player types?

K-means groups players with similar numbers.

- 6,095 attacker-seasons (≥450 min)
- Inputs (standardised): age, minutes, goals/90, assists/90
- Not used: market value, league
- k = 2 to 6 tested; k = 4 kept for readability



0.21–0.22

silhouette at every k, no sharp elbow: weak separation. Four interpretable but overlapping profiles.

€8M
Regular high-minute starters

€5M
Creative attackers

€1.5M
Younger, lower output

€1M
Older, fewer minutes

What the analysis shows

Age

Value rises into the early/mid twenties and falls after 30: a hump, not a line.

Playing time

One of the strongest simple associations (Spearman 0.51); causal direction is unclear.

Position

Performance statistics mean different things by role; goals matter most for attackers.

League

€13M Premier League median; top-five €4.5M vs others €600K.

Age × output

Young high-output attackers: highest median (€10M).

Clustering

K-means gives readable but weakly separated attacker profiles.

All associations: none of these shows that a characteristic causes a higher value.

What this project cannot show

Estimates

Transfermarkt values are estimates, not prices

No causation

Associations only; factors overlap

Missing roles

No defensive or goalkeeper statistics

No injuries

Injury history is not in the data

League and club

Context overlaps; club strength not measured

Repeated players

The same player appears in up to five seasons

Thresholds

450 min, July 31, 365 days are practical choices

Russia / Ukraine

War conditions after 2022 affect those leagues